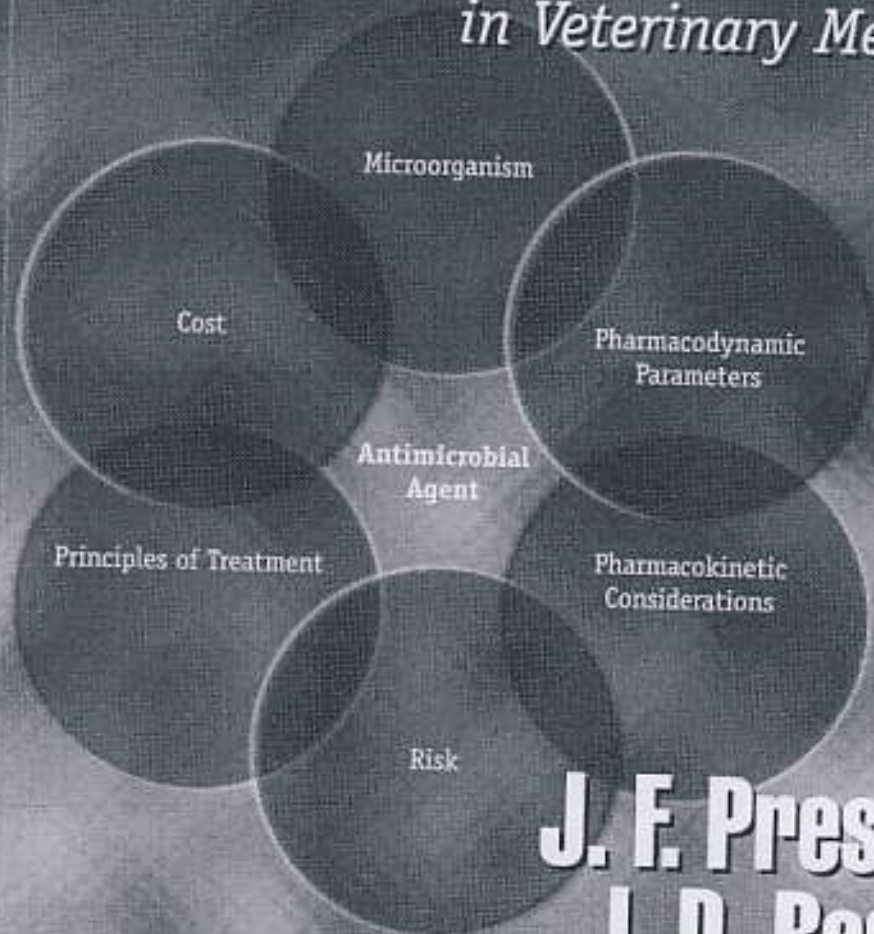


Third Edition

Antimicrobial Therapy

in Veterinary Medicine



J. F. Prescott
J. D. Baggot
R. D. Walker

Antimicrobial Therapy in Veterinary Medicine

THIRD EDITION

Edited by

John F. Prescott

M.A., Vet.M.B., Ph.D.

J. Desmond Baggot

M.V.M., Ph.D., D.Sc., F.R.C.V.S.

Robert D. Walker

M.Sc., Ph.D.

with Thirty Contributors



Iowa State Press

A Blackwell Publishing Company

John F. Prescott, M.A., Vet.M.B., Ph.D., is Professor in the Department of Pathobiology, University of Guelph, Ontario, Canada.

J. Desmond Baggot, M.V.M., Ph.D., D.Sc., F.R.C.V.S., is formerly Professor of Preclinical Veterinary Studies, Faculty of Veterinary Science, University of Zimbabwe, Harare, Zimbabwe, and currently Visiting Professor of Comparative Pharmacology, Victorian College of Pharmacy, Monash University, Melbourne, Australia.

Robert D. Walker, M.Sc., Ph.D., is formerly Professor in the Animal Health Diagnostic Laboratory, College of Veterinary Medicine, Michigan State University, East Lansing, and currently Director, Division of Animal and Food Microbiology, Center for Veterinary Medicine, U.S. Food and Drug Administration.

© 2000, 1993 Iowa State University Press; © 1988 Blackwell Scientific Publications

All rights reserved

Iowa State Press
A Blackwell Publishing Company
2121 State Avenue, Ames, Iowa 50014
Orders: 1-800-862-6037
Office: 1-515-292-0140
Fax: 1-515-292-3348
Web site: www.iowastatepress.com

© Printed on acid-free paper in the United States of America

Authorization to photocopy items for internal or personal use, or the internal or personal use of specific clients, is granted by Iowa State Press, provided that the base fee of \$10 per copy is paid directly to the Copyright Clearance Center, 222 Rosewood Drive, Danvers, MA 01923. For those organizations that have been granted a photocopy license by CCC, a separate system of payments has been arranged. The fee code for users of the Transactional Reporting Service is 0-8138-0779-4/2000 \$10.

First edition, 1988
Second edition, 1993
Third edition, 2000

Library of Congress Cataloging-in-Publication Data

Antimicrobial therapy in veterinary medicine / edited by John F. Prescott, J. Desmond Baggot, R.D. Walker; with thirty contributors.—3rd ed.

p. cm

Includes bibliographical references.

ISBN 0-8138-0779-4

1. Anti-infective agents in veterinary medicine. 2. Communicable diseases in animals—Chemotherapy. I. Prescott, John F. (John Francis) II. Baggot, J. Desmond. III. Walker, R.D. (Robert D.)

SP918.A45 A58 2000

636.0895329—dc21

The last digit is the print number: 9 8 7 6 5 4 3

00-036967

13

Tetracyclines

J. F. PRESCOTT

Tetracyclines are products of *Streptomyces* spp. (chlortetracycline, oxytetracycline, tetracycline) and their semisynthetic derivatives (doxycycline, minocycline). They are classic broad-spectrum antibiotics because of their activity against Gram-positive and Gram-negative aerobic and anaerobic bacteria, including *Brucella*, *Coxiella*, *Ehrlichia*, some *Mycobacterium*, *Mycoplasma*, and *Rickettsia*. The extraordinarily widespread nature of acquired resistance that has developed in the many common pathogens and particularly Gram-negatives limits the use of tetracyclines in many clinical circumstances. They have generally favorable pharmacokinetic properties, particularly the more lipophilic semisynthetic derivatives, but IM administration is painful, and they produce gastrointestinal disturbances if given orally. They have a number of other uncommon but important adverse effects. They tend to be used as first-line, workhorse antibiotics in ruminants and swine but have some very specific uses in companion animal settings in the treatment of chlamydiosis, coxiellosis, and rickettsiosis, and for some mycobacterial and mycoplasmal infections. Newer derivatives resistant to common resistance mechanisms are on the horizon.

Chemistry

The tetracyclines (chlortetracycline, oxytetracycline, tetracycline, doxycycline, minocycline) are crystalline amphoteric substances that are only slightly soluble in water at pH 7.0. They are available for use, mainly as the hydrochlorides, in a wide variety of dosage forms, both oral and parenteral. Solutions of the hydrochlorides are acidic and, with the exception of

TABLE 13.3. Usual oral dosages of tetracyclines for monogastric animals

Tetracycline Preparation	Maintenance Dose ^a	Dosage Interval (h)
Tetracycline hydrochloride	15 mg/kg	6-8
Swine	200-800 ppm	
Oxytetracycline hydrochloride	20 mg/kg	8-12
Minocycline hydrochloride	5 mg/kg	12
Doxycycline hyclate	5 mg/kg	12
Swine	200-250 ppm	

^aOne or preferably 2 priming doses should be double this amount.

PO to dogs and cats. Milk, antacids, and ferrous sulfate interfere with absorption. A generally satisfactory approach to PO administration is to initiate therapy with 1 or preferably 2 priming doses and to continue with maintenance doses (each one-half the priming dose) at 12-hour intervals. In the case of tetracycline hydrochloride, a 6-hour interval is recommended (Table 13.3).

Because of poor water solubility, oxytetracycline dihydrate must be given in much higher doses (50 mg/kg) than the hydrochloride to produce equivalent tissue concentrations. In horses, the only route by which the parenteral preparation of oxytetracycline hydrochloride should be administered is by slow (5-minute) IV injection. The regimen that could be expected to maintain therapeutic plasma concentrations (1-5 µg/ml) consists of a priming dose of 5 mg/kg followed by maintenance doses of 3 mg/kg at 12-hour intervals. Because of biliary excretion and passive diffusion into the large bowel, tetracyclines may cause disturbance of the microbial flora of the colon and predispose to salmonellosis or "colitis X." The more lipophilic members of the group (minocycline, doxycycline) are even more likely to have this effect.

Intramuscular injection of tetracyclines cannot be recommended for horses or companion animals because of local tissue damage, local pain, and erratic absorption. The recommended dose in cattle is 10 mg/kg given IM or preferably IV, because of variability in absorption. The long-acting oxytetracycline parenteral preparation, which is oxytetracycline base in 2-pyrrolidone, is approved for IM use in cattle and swine only. A single IM dose of 20 mg/kg provides serum concentrations of oxytetracycline above 0.5 µg/ml for 48 hours but appears to offer no advantage over the same dose of the conventional drug IM (Nouws, 1986). Subcutaneous injection in cattle maintains serum concentrations similar to those following IM administration and appears to be better tolerated. To prevent adverse effects, it is important to differentiate between the conventional and the long-acting formulations in dosage decisions.

Clinical Applications

The primary indications for tetracyclines are in the treatment of borreliosis, brucellosis, chlamydiosis, ehrlichiosis, leptospirosis, listeriosis, rickettsiosis, and tularemia. The older tetracyclines have been used for many years in managing infectious diseases in food animals because of their low cost, broad antimicrobial activity, ease of administration, and general effectiveness. Their widespread use has undoubtedly contributed to resistance in important pathogenic bacteria, which now limits the value of these drugs. The capacity of tetracyclines to attain effective concentrations in most tissues, except those separated by specialized cellular barriers, together with their broad spectrum of activity, makes them particularly useful in the treatment of mixed bacterial infections. The activity of the agents against rickettsias, chlamydiae, *Ehrlichia*, and some mycoplasmas makes them the drugs of choice in treatment of infections caused by these microorganisms. Although recommended for the treatment of plague, results in the treatment of experimental infections in animals have sometimes been disappointing. The tissue irritant effect of these drugs and their potential to cause colonic disturbance largely limit their use in horses. The lipophilic character of the newer tetracyclines (minocycline, doxycycline) allows them to attain concentrations in sites such as the prostate, which is poorly accessible to older members of the group. One disadvantage of tetracyclines over a number of other antimicrobial drugs is their bacteriostatic action, so that treatment may need to be for longer than with bactericidal drugs.

Tetracyclines, particularly doxycycline and minocycline, have anti-inflammatory properties independent of their antibacterial action. Experimentally, tetracyclines have protected mice from endotoxin-induced shock by reducing inflammatory cytokine and nitric oxide production.

Tetracyclines are commonly used in the treatment of brucellosis, usually in combination with rifampin or streptomycin. Doxycycline and minocycline are more effective than older tetracyclines because of better penetration into cells. Treatment with doxycycline or minocycline should last 6 weeks and with streptomycin 7–14 days. Tetracyclines (particularly minocycline and doxycycline) are also used in the treatment of infections caused by other intracellular bacteria, including *Ehrlichia* and *Coxiella*.

CATTLE, SHEEP, AND GOATS

Many of the microorganisms that cause bovine pneumonia are susceptible to tetracyclines at concentrations that can be achieved in lung tissue. The drugs are generally useful in the treatment of bovine pneumonias and also in their prophylaxis, especially in feedlots. Nevertheless, increasing resistance in *Pasteurella* and variable susceptibility of *Mycoplasma bovis* limit their effectiveness. The long-acting parenteral

tered orally for 3 weeks achieved complete remission of about half of canine patients with superficial pyoderma and partial remission in another 40%, but complete remission in only 14% of patients with deep pyoderma and partial remission in another 51% (Bettenay et al., 1998).

Cats suffering from chlamydial infection of the upper respiratory tract and conjunctiva should be treated with tetracyclines for 14 days to eliminate the organism and to remove the latent carrier state. Tetracyclines are drugs of choice for the treatment of *Haemobartonella felis*. Prolonged oral treatment with doxycycline does not eliminate the carrier state in *Bartonella henselae* or *B. clarridgeiae* infection (Kordick et al., 1997). Treatment by tetracyclines of a cat with *Yersinia pestis* infection was only temporarily effective (Culver, 1987).

POULTRY

Tetracyclines are effective in the treatment of chlamydiosis if administered for prolonged periods. Tetracycline or chlortetracycline can be administered in 1% medicated feed (45 days), and doxycycline has been administered at 100 mg/kg IM at 5-day intervals on 6 or 7 occasions (Gyldorff, 1987) or orally twice daily for 20 days. Tetracyclines are also used in the treatment of chronic respiratory disease (*Mycoplasma gallisepticum*) and infectious synovitis (*M. synoviae*), as well as of fowl cholera (*Pasteurella multocida*). Prolonged administration of oxytetracycline (250 ppm) in feed is required to control *M. gallisepticum* infection in birds. One report noted the surprising efficacy of tetracycline sorbate in the oral treatment of naturally occurring *Aspergillus fumigatus* infection (Roy et al., 1991).

Bibliography

- Anderson WL. 1983. Hemoglobinuria in cattle given long-acting oxytetracycline. *Mod Vet Pract* 64:997.
- Appleyard WT, Gilmour NJL. 1990. Use of long-acting oxytetracycline against pasteurellosis in lambs. *Vet Rec* 126:231.
- Ball HJ, Campbell JN. 1989. Antibiotic treatment of experimental *Mycoplasma californicum* mastitis. *Vet Rec* 125:377.
- Bettenay SV, et al. 1998. Doxycycline hydrochloride in treatment of canine pyoderma. *Aust Vet Pract* 28:14.
- Bousquet E, et al. 1998. Efficacy of doxycycline in feed for the control of pneumonia caused by *Pasteurella multocida* and *Mycoplasma hyopneumoniae* in fattening pigs. *Vet Rec* 143:269.
- Breitschwerdt EB, et al. 1998. Doxycycline hyclate treatment of experimental canine ehrlichiosis followed by challenge inoculation with two *Ehrlichia canis* strains. *Antimicrob Agents Chemother* 42:362.
- Brodie TA, et al. 1986. Some aspects of tick-borne diseases of British sheep. *Vet Rec* 118:415.
- Brodie TA, et al. 1988. Prophylactic use of long-acting tetracycline against tick-borne fever (*Cytoecetes phagocytophila*) in sheep. *Vet Rec* 122:43.
- Chaslus-Dancla E, et al. 1995. Tetracycline resistance determinants, Tet B and Tet M, detected in *Pasteurella haemolytica* and *Pasteurella multocida* from bovine herds. *J Antimicrob Chemother* 36:515.

- Chimo RS, et al. 1989. Long-acting oxytetracycline prophylaxis to protect susceptible cattle introduced into an area of Kenya with endemic East Coast fever. *Vet Rec* 124:219.
- Cook WR. 1973. Diarrhea in the horse associated with stress and tetracycline therapy. *Vet Rec* 93:15.
- Cranwell MP. 1990. Efficacy of long-acting oxytetracycline for the prevention of tick-borne fever in calves. *Vet Rec* 126:334.
- Culver M. 1987. Treatment of bubonic plague in a cat. *J Am Vet Med Assoc* 191:1528.
- Dargatz DA, et al. 1990. Antimicrobial therapy for rams with *Brucella ovis* infection of the urogenital tract. *J Am Vet Med Assoc* 196:605.
- Flores-Castro R, Carmichael LE. 1978. Canine brucellosis: current status of methods for diagnosis and treatment. *Gaines Vet Symp* 27:17.
- Gallo GE, Berg JL. 1995. Efficacy of a feed-additive antibacterial combination for improving feedlot cattle performance and health. *Can Vet J* 36:223.
- George L, et al. 1988. Topically applied furazolidone or parenterally administered oxytetracycline for the treatment of infectious bovine keratoconjunctivitis. *J Am Vet Med Assoc* 192:1415.
- Greig A, Linklater KA. 1985. Field studies on the efficacy of a long-acting preparation of oxytetracycline in controlling outbreaks of enzootic abortion of sheep. *Vet Rec* 117:627.
- Griffin DD, et al. 1979. Experimental oxytetracycline toxicity in feedlot heifers. *Bovine Pract* 14:37.
- Gregorio-Thomas R, et al. 1994. The use of long-acting oxytetracycline for the treatment of ovine footrot. *Br Vet J* 150:561.
- Guichon PT, et al. 1993. Comparison of two formulations of oxytetracycline given prophylactically to reduce the incidence of bovine respiratory disease in feedlot calves. *Can Vet J* 34:736.
- Gyldorff I. 1987. The treatment of chlamydiosis in psittacine birds. *Isr J Vet Med* 43:11.
- Gyrd-Hansen N, et al. 1981. Cardiovascular effects of intravenous administration of tetracycline in cattle. *J Vet Pharmacol Ther* 6:15.
- Hayashi K, et al. 1998. Clinical and microbiological effects of controlled-release local delivery of minocycline on periodontitis in dogs. *Am J Vet Res* 59:464.
- Hosie BD. 1988. Keratoconjunctivitis in a hill sheep flock. *Vet Rec* 122:40.
- Hosie BD, Greig A. 1995. Role of oxytetracycline dihydrate in the treatment of *Mycoplasma*-associated ovine keratoconjunctivitis in lambs. *Br Vet J* 151:83.
- Jordan D, Venning CM. 1995. Treatment of ovine dermatophilosis with long-acting oxytetracycline or a lincomycin-spectinomycin combination. *Aust Vet J* 72:234.
- Kordick DL, et al. 1997. Efficacy of enrofloxacin or doxycycline for treatment of *Bartonella henselae* or *Bartonella clarridgeiae* infection in cats. *Antimicrob Agents Chemother* 41:2448.
- Lairmore MD, et al. 1984. Oxytetracycline-associated nephrotoxicosis in feedlot cattle. *J Am Vet Med Assoc* 185:793.
- Ling GV, et al. 1980. Urine concentrations of chloramphenicol, tetracycline, and sulfisoxazole after oral administration to healthy adult dogs. *Am J Vet Res* 41:950.
- Love RJ, Love DN. 1977. Control of proliferative haemorrhagic enteropathy in pigs. *Vet Rec* 100:473.
- Madigan JE, Gribble D. 1987. Equine ehrlichiosis in northern California: 49 cases (1968-1981). *J Am Vet Med Assoc* 190:445.
- Marin CM, et al. 1989. Efficacy of long-acting oxytetracycline alone or in combination with streptomycin for treatment of *Brucella ovis* infection of rams. *Am J Vet Res* 50:560.
- Martin SW, et al. 1982. Factors associated with mortality and treatment costs in feedlot calves: the Bruce County beef project, years 1978, 1979, 1980. *Can J Comp Med* 46:341.
- Mebus CA, Logan LL. 1988. Heartwater disease of domestic and wild ruminants. *J Am Vet Med Assoc* 192:950.
- Nicoletti P, Chase A. 1987. The use of antibiotics to control canine brucellosis. *Compend Contin Educ Pract Vet* 9:1063.

- Nicoletti P, et al. 1985. Efficacy of long-acting oxytetracycline alone or in combination with streptomycin in the treatment of bovine brucellosis. *J Am Vet Med Assoc* 187:493.
- Nielsen P, Gyrd-Hansen N. 1996. Bioavailability of oxytetracycline, tetracycline, and chlortetracycline after oral administration to fed and fasted pigs. *J Vet Pharmacol Ther* 19:305.
- Nouws JFM. 1984. Irritation, bioavailability, and residue aspects of ten oxytetracycline formulations administered intramuscularly to pigs. *Vet Q* 6:80.
- Nouws JFM. 1986. Factors affecting the oxytetracycline disposition kinetics in ruminants—a review. *Ir Vet News* (May):9.
- Nouws JFM, et al. 1990. A comparative study on irritation and residue aspects of five oxytetracycline formulations administered intramuscularly to calves, pigs and sheep. *Vet Q* 12:129.
- Palmer JE. 1989. Prevention of Potomac horse fever. *Cornell Vet* 79:201.
- Potter WL. 1973. Collapse following intravenous injection of oxytetracycline in two horses. *Aust Vet J* 49:547.
- Radwan AI, et al. 1989. Experimental treatment of *Brucella melitensis* infection in sheep with oxytetracycline alone or combined with streptomycin. *Trop Anim Health Prod* 21:211.
- Riond J-L, Riviere JE. 1989. Effects of tetracyclines on the kidney in cattle and dogs. *J Am Vet Med Assoc* 195:995.
- Riond J-L, Riviere JE. 1990. Pharmacokinetics and metabolic inertness of doxycycline in young pigs. *Am J Vet Res* 51:1271.
- Riond J-L, et al. 1990. Comparative pharmacokinetics of doxycycline in cats and dogs. *J Vet Pharmacol Ther* 13:415.
- Roy S, et al. 1991. Use of tetracycline sorbate for the treatment of *Aspergillus fumigatus* infection in broiler chickens. *Br Vet J* 147:549.
- Shaw DH, Rubin SL. 1986. Pharmacologic activity of doxycycline. *J Am Vet Med Assoc* 189:508.
- Shurter JK, Elliott JB. 1998. Papillomatous digital dermatitis: treatment and control strategies—Part I. *Compend Cont Educ Pract Vet* 20:5158.
- Simpson BC, et al. 1987. Protection of cattle against heartwater in Botswana: comparative efficacy of different methods against natural and blood-derived challenges. *Vet Rec* 120:133.
- Stahlheim OHV. 1967. Chemotherapy of renal leptospirosis in swine. *Am J Vet Res* 28:161.
- Stevenson S. 1980. Oxytetracycline nephrotoxicosis in two dogs. *J Am Vet Med Assoc* 176:530.
- Swift BL, Thomas GM. 1983. Bovine anaplasmosis: elimination of the carrier state with injectable long-acting oxytetracycline. *J Am Vet Med Assoc* 183:63.
- Taylor SM, et al. 1986. Inhibition of *Babesia divergens* in cattle by oxytetracycline. *Vet Rec* 118:98.
- Vaala WE, et al. 1987. Acute renal failure associated with administration of excessive amounts of tetracycline in a cow. *J Am Vet Med Assoc* 191:1601.
- Van Donkersgoed J. 1992. Meta-analysis of field trials of antimicrobial mass medication for prophylaxis of bovine respiratory disease in feedlot cattle. *Can Vet J* 33:786.
- Wilson AW, et al. 1979. Chemotherapy of acute bovine anaplasmosis. *Aust Vet J* 55:71.
- Wilson RC, et al. 1985. Compartmental and noncompartmental pharmacokinetic analyses of minocycline hydrochloride in the dog. *Am J Vet Res* 46:1315.